

ECO 209 Assignment 1

Name: _____

1. (20 Points) Suppose a country's economy can be fully described by the following activities:

- A gold mining company pays its workers \$2800 to mine and process 100 kg of gold. The company sells 80 kg to a decoration manufacturer and forges 10 gold bars, which are sold directly to consumers for \$5000.
- A decoration manufacturer buys 80kg of gold at \$100 per kg and pays its workforce \$21,000 to produce 7 gold decorations. Out of these, 4 are sold to end consumers, 1 to a government-run museum, and 2 are exported for \$12,000 each.

(a) (12 pts) Calculate the GDP in the economy using all three methods. Please show your work in detail.

(b) (3 pts) Suppose that in the following year, the amounts of gold mined and gold decorations produced remain the same, wages paid by both firms remain the same, but the price of gold is now \$110 per kg, and the price of a decoration is \$15,000. What is the growth rate of real GDP between the first year and the next? Explain your answer.

(c) (5 pts) Using the GDP deflator method, calculate the percentage change in the general price level across the two years. Show your work.

Solution:

- (a) Using the income approach, we need to sum all the labour income and capital income. The labour income consists of \$2,800 from the gold mining company and \$21,000 from the decoration manufacturer. The business owner's earnings are calculated as revenue minus costs.

For the gold mining company, the owner earns \$100 (the price per unit) multiplied by 80 (units produced), which is \$8,000 in revenue. Also, the owner earn \$5000 from selling gold bar directly to consumer. Subtracting the labour cost of \$2,800, the owner's earnings are \$10,200.

For the decoration manufacturer, the owner earns 7 multiplied by \$12,000 (the price per unit) (units produced) minus the costs (\$21,000 for labour and \$8,000 for materials), resulting in earnings of \$55,000.

Summing all the labour and capital income:

$$GDP = \$2,800 + \$10,200 + \$21,000 + \$55,000 = \$89,000$$

Using the expenditure approach, we need to sum all the final goods consumption, which is:

$$GDP = C + G + NX = \$5000 + 4 \times \$12,000 + 1 \times \$12,000 + 2 \times \$12,000 = \$89,000$$

Using the value-added approach, we first calculate the value added for each firm. For the gold mining company, it doesn't incur any intermediate costs, so its value added is the revenue, which is \$13,000. For the decoration manufacturer, its

value added is \$84,000 (revenue) minus \$8,000, resulting in \$76,000.

$$GDP = \$13,000 + \$76,000 = \$89,000$$

(b) Real GDP remains unchanged as production quantities remain constant. The only variable that changes is the price. Therefore, the growth rate is zero.

(c) First, use any method to find the nominal GDP in the next year = \$5000 + \$15000 * 7 = \$110,000

Now, find the GDP Deflator = NGDP/RGDP = 110,000/89,000 = 1.24

Percentage change in prices = (1.24-1)*100% = 24%

2. (10 Points) Suppose x , y , and z grow at constant rates given by $\bar{x}$, $\bar{y}$ and $\bar{z}$. What is the growth rate of z in each of the following cases?

(a) (2 pts) $z = xy$

(b) (4 pts) $z = x^{-0.4} \left(\frac{1}{y} \right)^{0.2}$

(c) (4 pts) $z = \frac{y^{0.3}}{x^{0.2}}$

Solution:

(a) $g_z = g_x + g_y$

(b) $g_z = g(x^{-0.4}) + g\left(\left(\frac{1}{y}\right)^{0.2}\right) = -0.4g_x + 0.2g(1/y) = -0.4g_x + 0.2g(y^{-1}) = -0.4g_x - 0.2g_y$

(c) $z = y^{0.3}x^{-0.2}, g_z = 0.3g_y - 0.2g_x$

3. (10 Points) Suppose that country A is growing at 3% per year, and country B is growing at 2%. In the year 2000, the GDP per capita of country A is \$8,000, and the GDP per capita of country B is \$30,000.

(a) (5 pts) How long would it take country A to catch up with country B in term of GDP per capita?

(b) (5 pts) With the help of foreign investment and technology diffusion, country A can grow at 5% in the initial 20 years. How long would it take for country A to catch up with country B in this case? (Hint: If a country is growing at 2% in the first 2 years and the growth rate falls to 1%, we can calculate the GDP after 5 years as $Y_5 = Y_0(1.02)^2(1.01)^3$.)

Solution:

- (a) To calculate that, we set the GDP in year t for country A and country B to be equal.

$$Y_t^A = Y_t^B$$

We can apply the constant growth rate formula

$$Y_0^A(1 + g_A)^t = Y_0^B(1 + g_B)^t$$

$$8000(1.03)^t = 30000(1.02)^t \rightarrow \left(\frac{1.03}{1.02} \right)^t = 3.75$$

$$\ln \left(\frac{1.03}{1.02} \right)^t = \ln(3.75) \rightarrow t \ln \left(\frac{1.03}{1.02} \right) = \ln(3.75)$$

$$t = 135.48$$

- (b) The procedure is similar except that for the first 20 years, the growth rate for country A is at 5%

$$8000(1.05)^{20}(1.03)^{t-20} = 30000(1.02)^t$$

$$\frac{(1.03)^t}{(1.03)^{20}(1.02)^t} = \frac{30000}{8000(1.05)^{20}} \rightarrow \left(\frac{1.03}{1.02} \right)^t = \frac{30000}{8000(1.05)^{20}}(1.03)^{20}$$

$$t \ln \left(\left(\frac{1.03}{1.02} \right) \right) = \ln \left(\frac{30000}{8000(1.05)^{20}}(1.03)^{20} \right)$$

$$t = 96.05$$

4. (15 Points) Do the following production functions exhibit increasing, constant, or decreasing returns to scale in both K (capital) and L (labour)? ($\bar{A}$ is a fixed positive number)

1. (5 pts) $Y = \bar{A}K^{1/3}L^{1/2}$

2. (5 pts) $Y = K^{1/3}L^{2/3} - \bar{A}$

3. (5 pts) $Y = L + K^{0.5}L^{0.5}$

Solution: When the production function is not Cobb-Douglas, we will check the return to scale by multiply the input by γ

(a) The sum of $\frac{1}{3} + \frac{1}{2} < 1$. Decreasing return to scale

(b) Although the sum of $1/3 + 2/3 = 1$, we are subtracting it by a constant term $\bar{A}$. We cannot tell what kind of return to scale by checking the power. Sub it $K = \gamma K$
 $L = \gamma L$

$$F(\gamma K, \gamma L) - \gamma F(K, L) = \gamma K^{1/3} L^{2/3} - \bar{A} - \gamma (K^{1/3} L^{2/3} - \bar{A})$$

$$\gamma K^{1/3} L^{2/3} - \bar{A} - \gamma K^{1/3} L^{2/3} + \gamma \bar{A} = (\gamma - 1)\bar{A}$$

So, the function is increasing return to scale

(c) $\gamma L + (\gamma K)^{0.5}(\gamma L)^{0.5} = \gamma L + \gamma(K)^{0.5}(L)^{0.5} = \gamma(L + K^{0.5}L^{0.5})$. Constant return to scale

5. (25 Points) This question relates to the Solow growth model discussed in Chapter 5. Mont Plaisant, a closed economy with no government spending, operates under the following production function:

$$Y_t = AK_t^{1/3}L_t^{2/3}$$

In this model, Y_t represents the output in period t , K_t denotes the capital stock in period t , and L_t stands for the labour employed in period t . The accumulation of capital follows the standard equation:

$$K_{t+1} = (1 - \delta)K_t + I_t$$

where δ is the depreciation rate and I_t is the investment in period t .

Mont Plaisant's population saves a fixed proportion, $\bar{s}$, of their output for investment each period. The population grows at a rate $\bar{n}$, following the equation $L_{t+1} = (1 + \bar{n})L_t$. Lowercase letters represent per-person values of the variables.

- (a) Convert the capital accumulation equation from aggregate to per-person terms. The left-hand side should equal Δk_{t+1} , while the right-hand side should include y_t , k_t , and the relevant parameters.
- (b) Find the steady-state values of capital per capita (k^*) and output per capita (y^*).
- (c) What happens to output per capita in the steady state when $\bar{s}$ increases? Use your findings from part (b) to explain.
- (d) After estimating the model, we have the following parameter values and exogenous variables: $\delta = 0.05$, $\bar{A} = 1$, $\bar{s} = 0.20$, and $\bar{n} = 0.02$. Additionally, the current capital per capita in period 1 is $k_1 = 4$. Using this information, calculate Δk_2 , Δk_3 , Δk_4 and Δk_5 . What pattern do you observe in Δk_t , and why does this pattern occur? Explain in 2-4 sentences.
- (e) How should we expect the growth rate of output in Mont Plaisant from period 1

onward? Explain in 2-4 sentences.

- (f) Draw a Solow diagram with k on the x-axis to illustrate the scenario described in part (d).

Solution:

(a) Plug in ($I_t = sY_t$)

$$K_{t+1} - K_t = \Delta K_{t+1} = sY_t - \delta K_t$$

Divide both side by K_t ,

$$\frac{\Delta K_{t+1}}{K_t} = g_K = s \frac{Y_t}{K_t} - \delta = s \frac{Y_t/L_t}{K_t/L_t} - \delta = s \frac{y_t}{k_t} - \delta$$

Now, recall that $k_t = \frac{K_t}{L_t}$ and apply the growth rate approximate formula:

$$g_k = g_K - g_L$$

Next, plug in the g_K above and the fact that $g_L = \bar{n}$

$$g_k = \frac{\Delta k_{t+1}}{k_t} = s \frac{y_t}{k_t} - \delta - \bar{n}$$

Multiply both sides by k_t

$$\Delta k_{t+1} = sy_t - (\delta + \bar{n})k_t$$

(b) In steady state, $\Delta k_{t+1} = 0$

$$0 = sy^* - (\delta + \bar{n})k^*$$

$$(\delta + \bar{n})k^* = sy^*$$

$$\frac{k^*}{y^*} = \frac{s}{\delta + \bar{n}}$$

Rearrange the equation to $k^* = \frac{s}{\delta + \bar{n}} y^*$ and plug it into the output function ($y_t =$

$Ak_t^{1/3}$):

$$y^* = A \left(\frac{s}{\delta + \bar{n}} y^* \right)^{1/3}$$

$$y^{*2/3} = A \left(\frac{s}{\delta + \bar{n}} \right)^{1/3}$$

$$y^* = A^{3/2} \left(\frac{s}{\delta + \bar{n}} \right)^{1/2}$$

The steady-state k^* is then

$$k^* = A^{3/2} \left(\frac{s}{\delta + \bar{n}} \right)^{3/2}$$

- (c) We can use the formula we derived earlier to calculate the change in capital per person.

$$\Delta k_{t+1} = sy_t - (\delta + \bar{n})k_t$$

$\Delta k_2 = 0.037$, $\Delta k_3 = 0.036$, $\Delta k_4 = 0.034$, and $\Delta k_5 = 0.031$.

We observe that the change in capital is positive, meaning capital increases each period, but at a decreasing rate. This occurs because the economy is currently below its steady state. But, as the economy approaches the steady state, the growth rate gradually declines.

- (d) Since the economy is currently below its steady state, the economy will experience positive growth in output. It is because there will be positive net investment (i.e., $sy_t > (\delta + \bar{n})k_t$), leading to capital growth in addition to population growth. The growth rate would slow down as the economy get closer to the steady state.
- (e)

